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Research Paper

Advancing biomedical waste classification through a hybrid ensemble of deep Learning, reinforcement Learning, and differential evolution algorithms ☆

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Highlight

- Novel AI waste classification model introduced.
- Uses ensemble deep learning techniques.
- Enhances accuracy and efficiency in waste management.
- Tested on diverse, real-world datasets.
- Potential to improve sustainability in healthcare environments.

Abstract

The complex nature of pharmaceutical and biomedical waste poses significant challenges for effective management, particularly in the safe and cost-intensive disposal of infectious materials. This research presents a novel classification model that utilizes a double heterogeneous ensemble integrating deep learning, reinforcement learning, and differential evolution algorithms for waste classification. The model operates through three key phases: image augmentation, ensemble image segmentation, and ensemble convolutional neural network architectures, employing decision fusion techniques that incorporate reinforcement learning and differential evolution. It integrates various image segmentation methods, including U-Net, Mask R-CNN, DeepLab Version 3 Plus, and convolutional neural network architectures such as Inception Version 3, Residual Network 50, Mobile Network Version 2, and Densely Connected Convolutional Network 121.The developed model powers the "Biosorter," a machine specifically designed to differentiate between infectious and non-infectious waste. Comprehensive evaluations conducted on both proprietary and benchmark datasets demonstrate that the proposed BioSorter model significantly outperforms several widely used deep learning architectures—including ResNet50, DenseNet121, MobileNetV2, InceptionV3, and ResNeXt-50. On average, the model achieved classification improvements of 5.35% and 9.05% in accuracy over these methods on the respective datasets. During real-world deployment at a small medical center, the BioSorter achieved 98% sorting accuracy and a 50% increase in processing throughput. Furthermore, a post-deployment usability assessment was conducted using the System Usability Scale (SUS)—a standardized questionnaire commonly used to evaluate perceived ease of use of interactive systems. The BioSorter received a score of 93.5 on the SUS (out of 100), reflecting a high level of user-perceived efficiency and interface simplicity during operational use. This study represents a significant advancement in waste management technology, offering potential to reduce disposal costs and enhance sustainability in healthcare environments.

Introduction

Effective management of waste from the pharmaceutical and biomedical sectors is a critical and complex task due to the diverse types of waste, including hazardous, recyclable, and general waste. Effective management of waste from the pharmaceutical and biomedical sectors is a critical and complex task due to the diverse types of waste, including hazardous, recyclable, and general waste. The heterogeneity of this waste stream necessitates a nuanced approach to ensure environmental safety and public health protection (Salman et al., 2016). The COVID-19 pandemic has further emphasized the urgency of this issue, leading to an increase in biomedical waste and heightening the risk of environmental contamination (Kanwal et al., 2022, Olaniyan et al., 2022).

Biomedical waste is significantly more expensive and hazardous to dispose of than recyclable or general waste. Therefore, reducing the volume of biomedical waste through immediate and accurate separation at the point of generation is both cost-effective and environmentally beneficial. This study addresses this challenge through the development of the BioSorter system, which automates waste identification and sorting to improve safety, efficiency, and sustainability.

The rapid evolution of machine learning (ML), deep learning (DL), and artificial intelligence (AI) has profoundly impacted waste management, offering innovative solutions to long-standing challenges. Recent research highlights AI-driven methods for waste classification and segregation that enhance resource optimization and reduce health hazards. For example, advanced convolutional neural networks (CNNs) have been implemented in automated waste segregation systems, achieving impressive classification accuracies of up to 99.15% in smart city projects (Azis et al., 2023).

Additionally, the integration of AI with Internet of Things technologies has revolutionized real-time monitoring and automated sorting processes for biomedical and solid wastes. These integrated systems have demonstrated significant improvements in operational efficiency while minimizing human intervention and associated errors (Zhang et al., 2023). Deep learning models, such as YOLOv8, have further advanced waste detection and classification, delivering robust performance in urban waste management scenarios (Bawankule et al., 2023).

The adoption of AI in waste management extends beyond operational efficiencies to support sustainability objectives, including enhanced recycling rates and reduced carbon emissions (Sharma and Vaid, 2021). These advancements underline the transformative potential of ML and DL technologies to address pressing issues in municipal and healthcare waste management, offering scalable, adaptable, and environmentally safe solutions.

Managing healthcare waste, particularly infectious waste, presents unique challenges distinct from other solid waste streams. Effective infection control, regulatory compliance, and practical implementation are critical aspects that must be addressed to ensure safe and efficient waste management. The complexity of accurately sorting infectious and non-infectious waste exacerbates the potential for human error, which can compromise safety and regulatory compliance. The volume and variety of waste generated in populous countries demand a comprehensive and sophisticated management strategy (Kodali et al., 2022). Current waste management systems often struggle to differentiate effectively between various waste types (Gupta et al., 2023), a limitation particularly concerning given the sustainability challenges posed by conventional disposal methods (Sapkota and Pariatamby, 2023). Despite advancements in waste management practices, there remains a significant gap in the accurate and efficient classification of infectious versus non-infectious waste, leading to potential health risks and environmental contamination.

Recent advancements in machine learning (ML), deep learning (DL), and artificial intelligence (AI) have played a transformative role in improving waste classification. These technologies have been successfully deployed to automate waste separation tasks, improve classification accuracy, and reduce operational costs in municipal and e-waste domains (Hu et al., 2021, Nnamoko et al., 2022, Oberoi, 2021). Their adaptability across various waste types highlights the potential for intelligent, scalable, and sustainable solutions in complex waste streams, such as those in healthcare environments (Shen, 2023).

This research introduces the BioSorter, a fully integrated AI-driven waste sorting system that combines heterogeneous ensemble deep learning techniques with real-time automation. Designed for deployment at the point of waste generation, the BioSorter performs end-to-end classification and physical segregation of pharmaceutical and biomedical waste. It accurately differentiates between recyclable, general, and infectious waste, reducing the risk of human error and enhancing compliance with healthcare safety standards. The system's architecture includes a multi-model classification framework as well as automated mechanical components, enabling safe and efficient waste handling in real-world clinical environments.

The BioSorter integrates multiple machine learning algorithms trained to classify diverse types of biomedical and pharmaceutical waste, particularly those indicative of infectious contamination. The system is designed to support automated classification while minimizing manual handling and promoting alignment with biomedical waste management protocols. To achieve efficient and accurate classification, the system incorporates several convolutional neural network (CNN) architectures—DenseNet121, MobileNetV2, ResNet50, and InceptionV3. These models were selected based on their demonstrated performance in biomedical image analysis and their suitability for real-time, resource-constrained environments (Benbriqa et al., 2023, Dahj et al., 2023, Sethanan et al., 2023). Their complementary strengths enable the system to balance computational efficiency with robust feature extraction, ensuring adaptability across diverse operational contexts.

Unlike prior research that primarily focuses on standalone classifiers or laboratory-based sorting prototypes, the BioSorter introduces a real-time, fully deployable AI-powered system that combines heterogeneous deep learning models with physical waste segregation mechanisms. The novelty lies in the ensemble integration of DenseNet121, MobileNetV2, ResNet50, and InceptionV3—optimally combined using a differential evolution strategy—which allows the system to maintain both high classification accuracy and computational efficiency across diverse waste types. This hybridization not only enhances performance in resource-constrained healthcare environments but also represents one of the first demonstrations of an end-to-end AI-powered solution for biomedical waste classification and physical sorting in clinical settings.

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Section snippets

Related literature

This section reviews the critical challenges in biomedical and pharmaceutical waste management and surveys the application of advanced computational methods to address them. We first establish the significant health and environmental risks posed by this waste stream, then discuss the emergence of Artificial Intelligence (AI) as a solution, thereby identifying the specific research gap that our proposed model aims to fill. …

Research methods

This study develops a deep learning-based waste classification model for the pharmaceutical and biomedical sectors. It focuses on precise categorization of hazardous and recyclable waste, integrating techniques like image augmentation and segmentation. The methodological steps include data collection, preprocessing, and model testing, aiming to improve waste management and reduce related environmental and health risks.

Fig. 1 illustrates the complete research pipeline from dataset collection …

Computational result and framework

The study employed two computational setups: model training on Google Collaboratory with an NVIDIA Tesla V100 GPU (16GB RAM), and performance evaluations using dual Intel Xeon-2.30GHz CPUs, 52GB RAM, and a Tesla K80 GPU (16GB GPU RAM). The experiments were categorized into three parts: identifying optimal method combinations, comparing the proposed approach against state-of-the-art techniques on the PBW dataset, and evaluating the method's efficacy on the dataset from Zhou et al. (2022) for …

Significance of the study

In the context of escalating environmental and health concerns associated with pharmaceutical and biomedical waste, our research introduces a cutting-edge deep learning model designed to revolutionize waste classification within these critical sectors. The need for advanced waste management solutions is urgent, given the increasing volume of hazardous waste generated, particularly during the COVID-19 pandemic. This situation exacerbates the challenges of waste management due to the influx of …

Conclusion and outlook

The complex nature of pharmaceutical and biomedical waste, comprising hazardous, recyclable, and general materials, presents a major challenge for waste management. To address this, we introduce a novel waste classification model that leverages heterogeneous ensemble deep learning techniques. Driven by the urgent need to reduce infectious waste disposal costs (often shouldered

by medical centers) and protect human and environmental health, this model offers a significant technological …

CRediT authorship contribution statement

Surajet Khonjun: Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Rapeepan Pitakaso:** Writing – original draft, Project administration, Formal analysis. **Thanatkij Srichok:** Writing – review & editing, Resources, Formal analysis, Conceptualization. **Sarayut Gonwirat:** Software. **Nawinda Vanichakulthada:** Validation, Data curation. **Peerawat Luesak:** Visualization, Validation. **Ganokgarn Jirasirilerd:** Validation, Data curation. …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

Special issue articles

References (43)

H. Benbriqa *et al.*
Performance of Heterogenous Neuro-Fuzzy Ensembles over Medical Datasets
Sci. Afr. (2023)

J.N.M. Dahj *et al.*
A Novel Heterogenous Ensemble Theory for Symmetric 5G Cells Segmentation: Intelligent RAN Analytics
International Journal of Intelligent Networks (2023)

K. Lin *et al.*
Data-Driven Models applying in Household Hazardous Waste: Amount Prediction and Classification in Shanghai
Ecotoxicol. Environ. Saf. (2023)

S.-M.-V. Ngoc *et al.*
COVID-19 and Environmental Health: a Systematic Analysis for the Global Burden of Biomedical Waste by this Epidemic
Case Stud. Chem. Environ. Eng. (2022)

B. Sapkota *et al.*

Pharmaceutical Waste Management System – are the Current Techniques Sustainable, Eco-Friendly and Circular? a Review
Waste Manag. (2023)

K. Sethanan *et al.*
Double AMIS-Ensemble Deep Learning for Skin Cancer Classification
Expert Syst. Appl. (2023)

S. Stiernström *et al.*
Evaluation of Frameworks for Ecotoxicological Hazard Classification of Waste
Waste Manag. (2016)

A. Agrachev *et al.*
Control on the Manifolds of Mappings with a View to the Deep Learning
J. Dyn. Control. Syst. (2021)

Arun Kumar S, Hubert Mary. L, Shaamili Varsa G V., Ulaganathan. M.S, L. Kannagi, P. Shyamala Bharathi, 2023. A Novel...

F.A. Azis *et al.*
The Development of an Automated Waste Segregator
IJIE (2023)

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